

2.2 Problem solving and programming	
How computers can be used to solve problems and programs can be written to solve them <i>(Learners will benefit from being able to program in a procedure/imperative language.)</i>	
2.2.1 Programming techniques	a) Programming constructs: sequence, iteration, branching. b) Recursion, how it can be used and compares to an iterative approach. c) Global and local variables. d) Modularity, functions and procedures, parameter passing by value and by reference. e) Use of an IDE to develop/debug a program.
2.2.2 Computational methods	a) Features that make a problem solvable by computational methods. b) Problem Recognition. c) Problem Decomposition. d) Use of divide and conquer. e) Use of abstraction. f) Learners should apply their knowledge of: • backtracking • data mining • heuristics • performance modelling • pipelining • visualisation to solve problems.